

RANSOMEWATCH: AI-Based Real-Time Ransomware Detection

Presented by Aayushman Baruah

**Under the guidance of
MR. SOHAM GOSWAMI
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**MCKV Institute of Engineering
243, G.T. Road (North), Liluah, Howrah-711204**

**(NAAC Accredited Grade- An Autonomous Institute affiliated to Maulana Abul Kalam
Azad University of Technology, West Bengal)**

Introduction to RANSOMEWATCH

Critical Threat

Ransomware is a paramount modern cybersecurity challenge.

Detection Gap

Traditional systems struggle to detect zero-day threats.

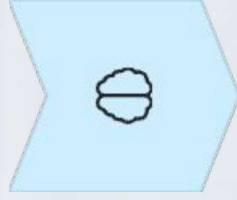
AI Solution

Machine learning enables dynamic behavior-based detection.

Hybrid System

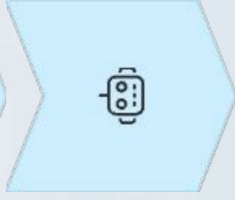
RANSOMEWATCH addresses these issues.

Literature Review: Foundational Concepts



ML in Cybersecurity

ML models identify malware patterns from network traffic.



Generative Adversarial Networks

GANs create synthetic data. This combats data scarcity.



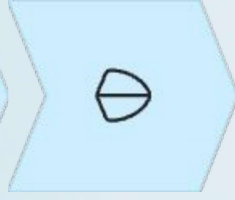
WGAN-GP Stability

WGAN-GP stabilises GAN training for quality samples.



Hybrid Model Advantages

Combining GAN data with classifiers improves accuracy.



RANSOMEWATCH Integration

RANSOMEWATCH integrates WGAN-GP and XGBoost for detection.

Motivation Behind

RaaS Proliferation

Rise of ransomware-as-a-service (RaaS) attacks demands new defenses.

Interpretable Tools

We aim for tools easily understood by non-technical users.

Real-time Accuracy

Need for accurate detection is critical.

Robust System

Our goal is a robust, interpretable UI.

System Architecture of RANSOMEWATCH

Modular design showing WCGAN, XGBoost

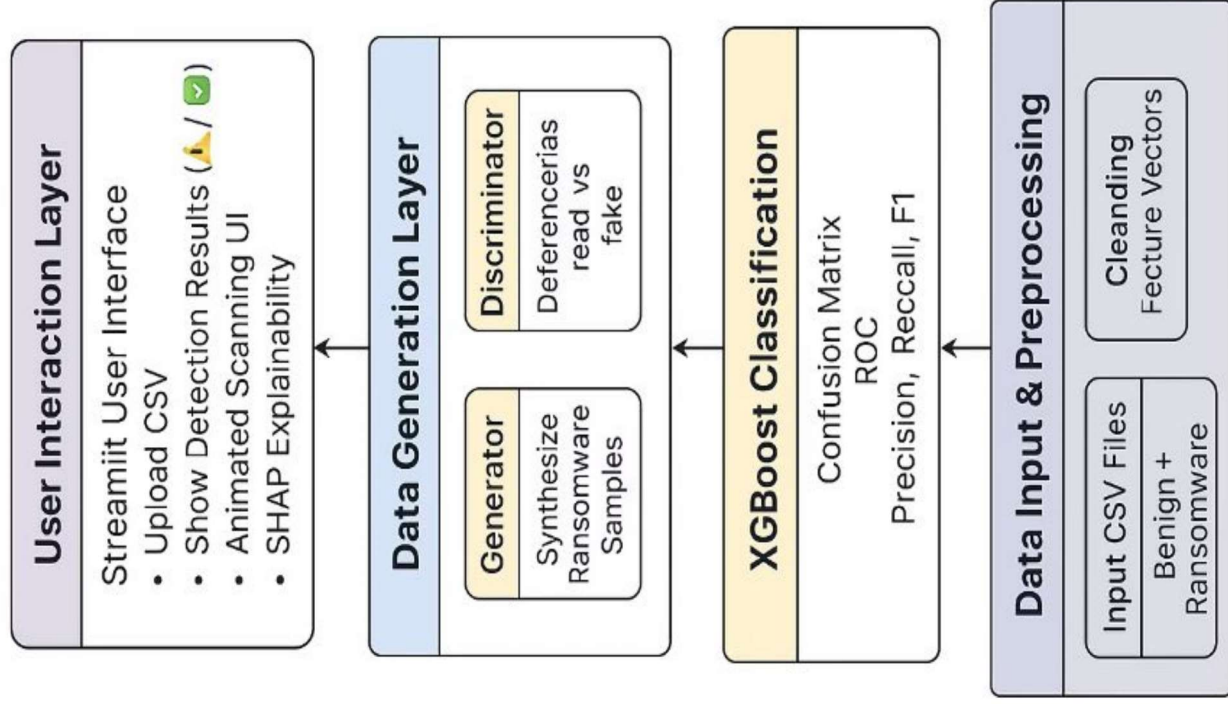


fig: A system architecture diagram illustrating data flow from collection through model training to user

RANSOMEWATCH Workflow



WCGAN Generation
Synthetic ransomware

XGBoost Classification
XGBoost classifier pro

Streamlit UI
Interactive user interfa

Datasets Utilised for Training

CICIDS 2017

Includes benign and various attack traffic samples.

Custom

Ransomware

Dataset features WannaCry, Ryuk, and other families.

Preprocessing

Steps

Involves feature selection, normalization, and encoding.

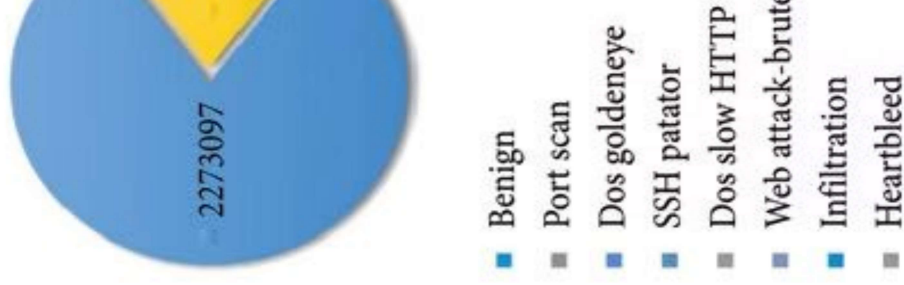


fig: Distribution

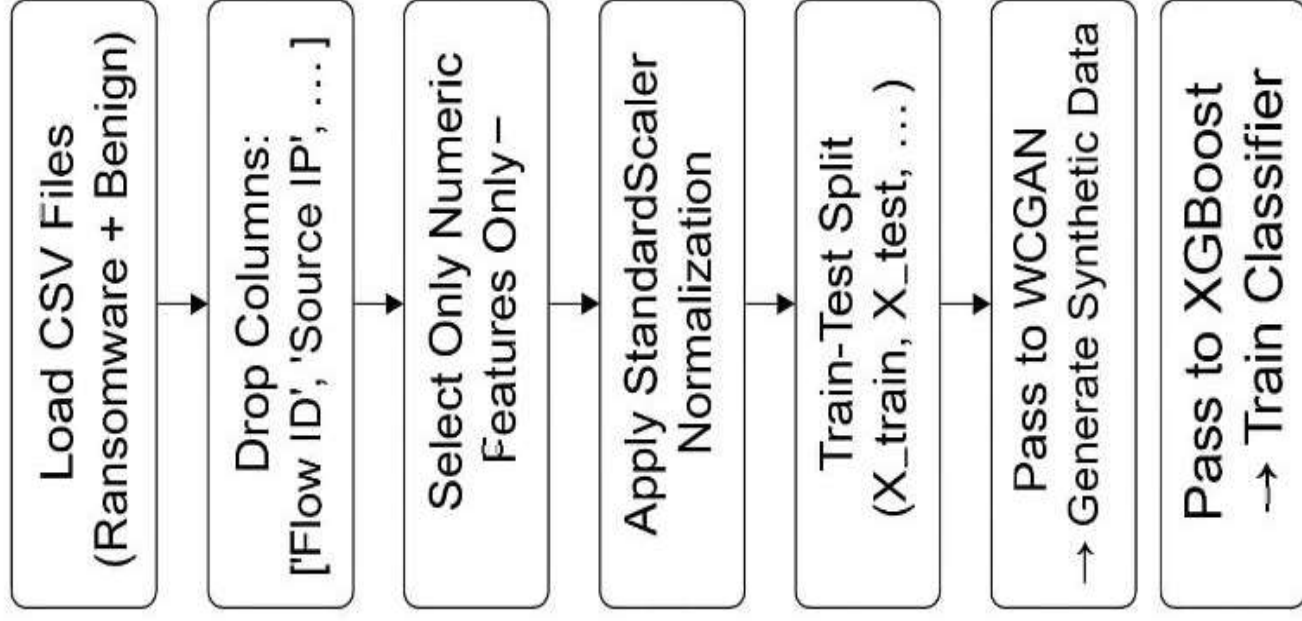


fig: Data Preprocessing Pipeline (Flowchart of feature selection, normalization, encoding)

Data Preprocess

The pipeline includes essential steps for preparing data for model training: **feature selection** to identify relevant attributes, **normalization** to ensure consistent range, and **encoding** to transform categorical variables into numerical format. This process ensures high-quality input for the model and facilitates anomaly detection.

WCGAN-GP Generator and Discriminator Architecture

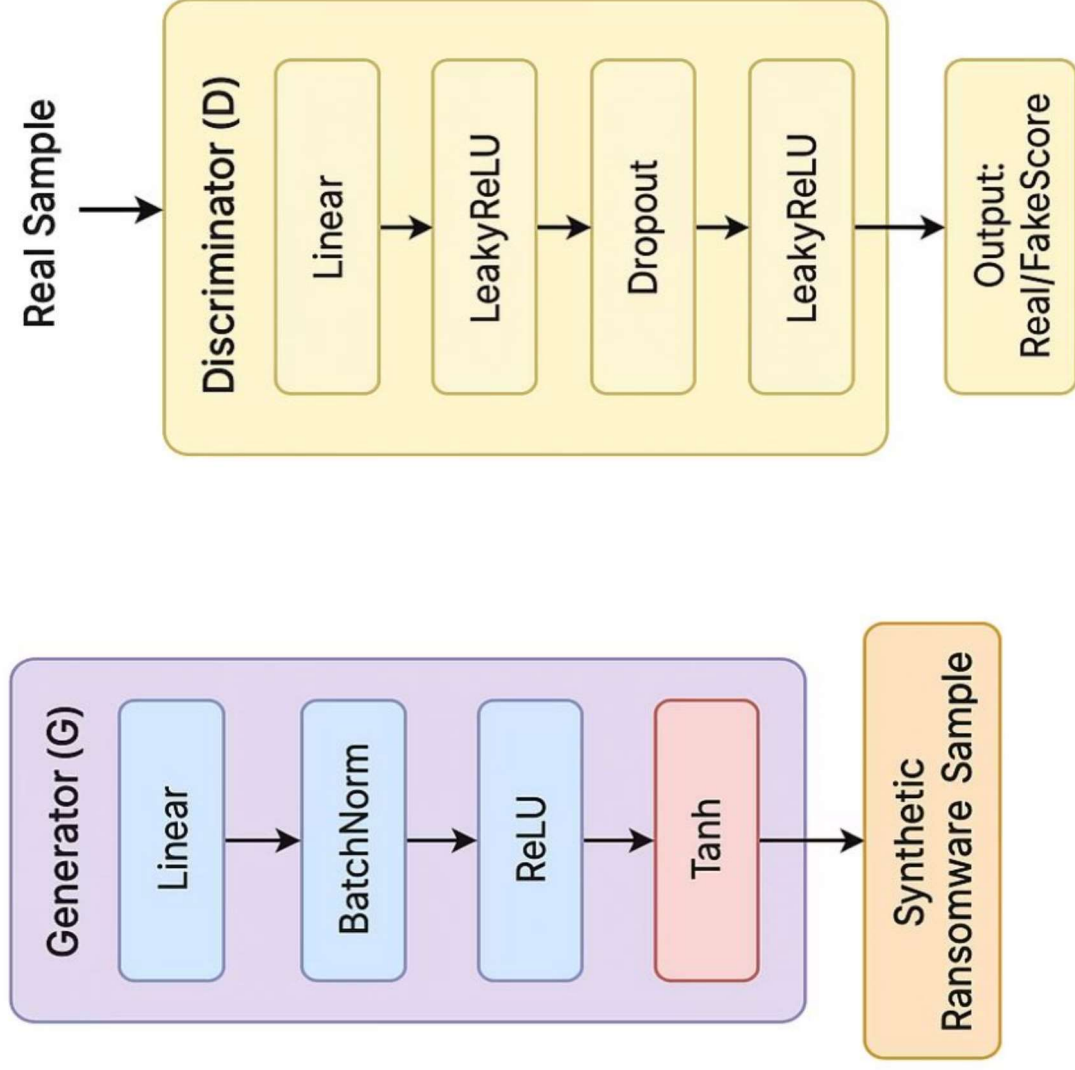
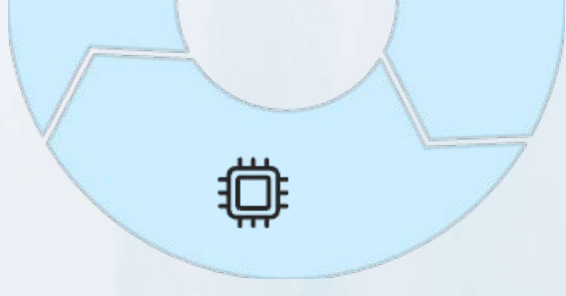


fig: Generator vs Discriminator Architecture in WCGAN-GP

Implemented

WCGAN-GP

Generates
high-quality
synthetic
ransomware
traffic.



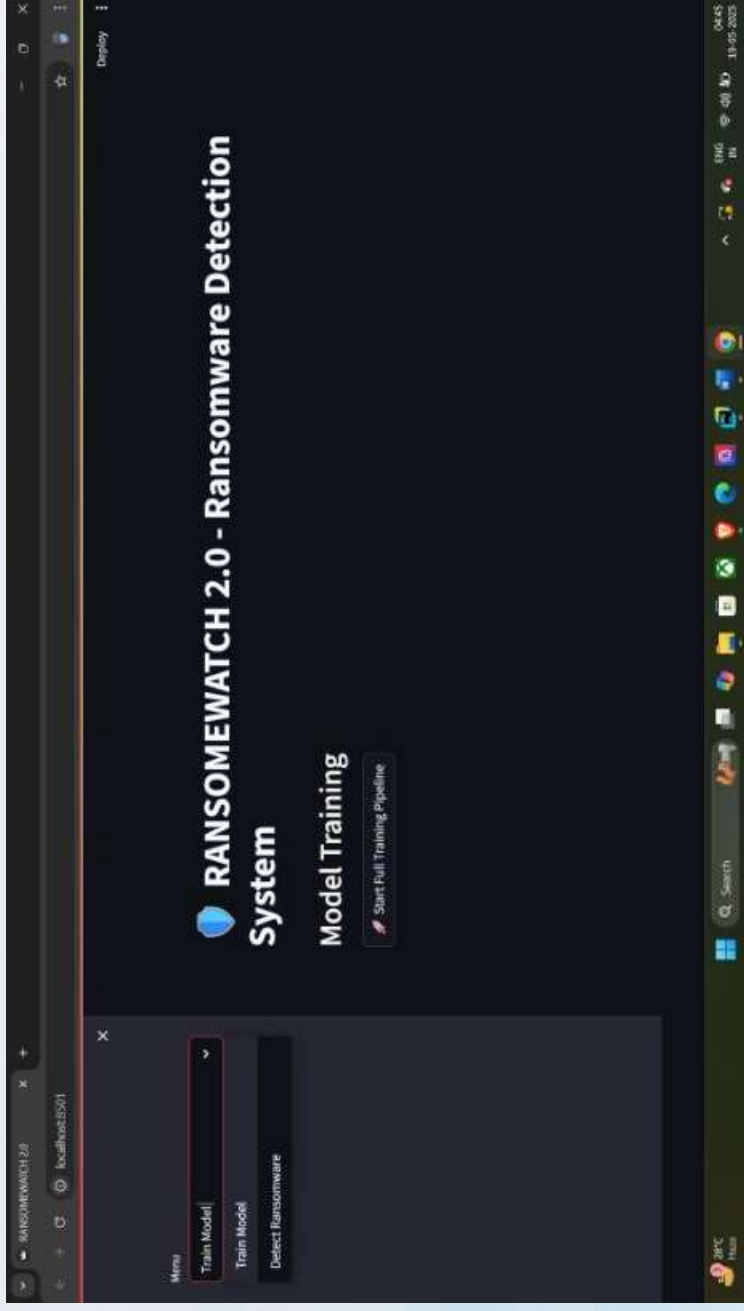


fig:Streamlit dashboard screenshot showing sidebar menu and main workspace for user interactions.

Streamlit Dash

This screenshot displays a sidebar menu for easy navigation and an interactive user interface.

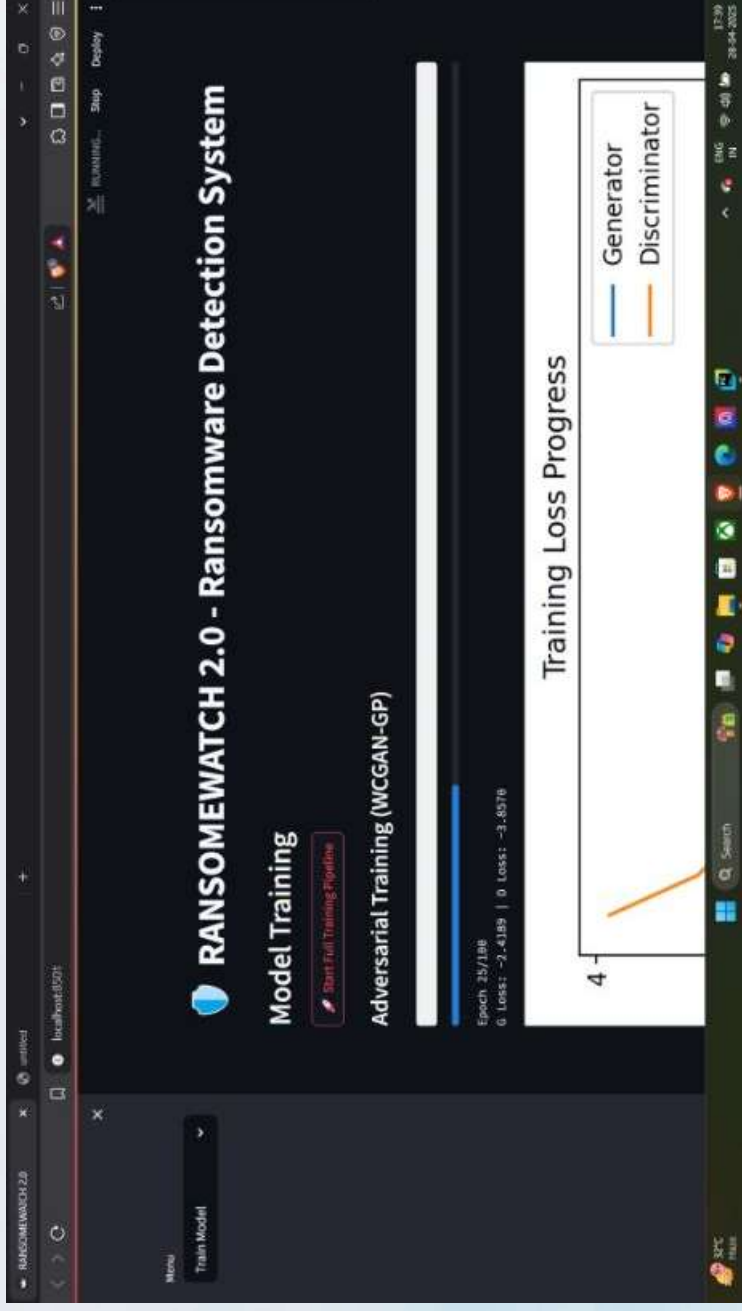
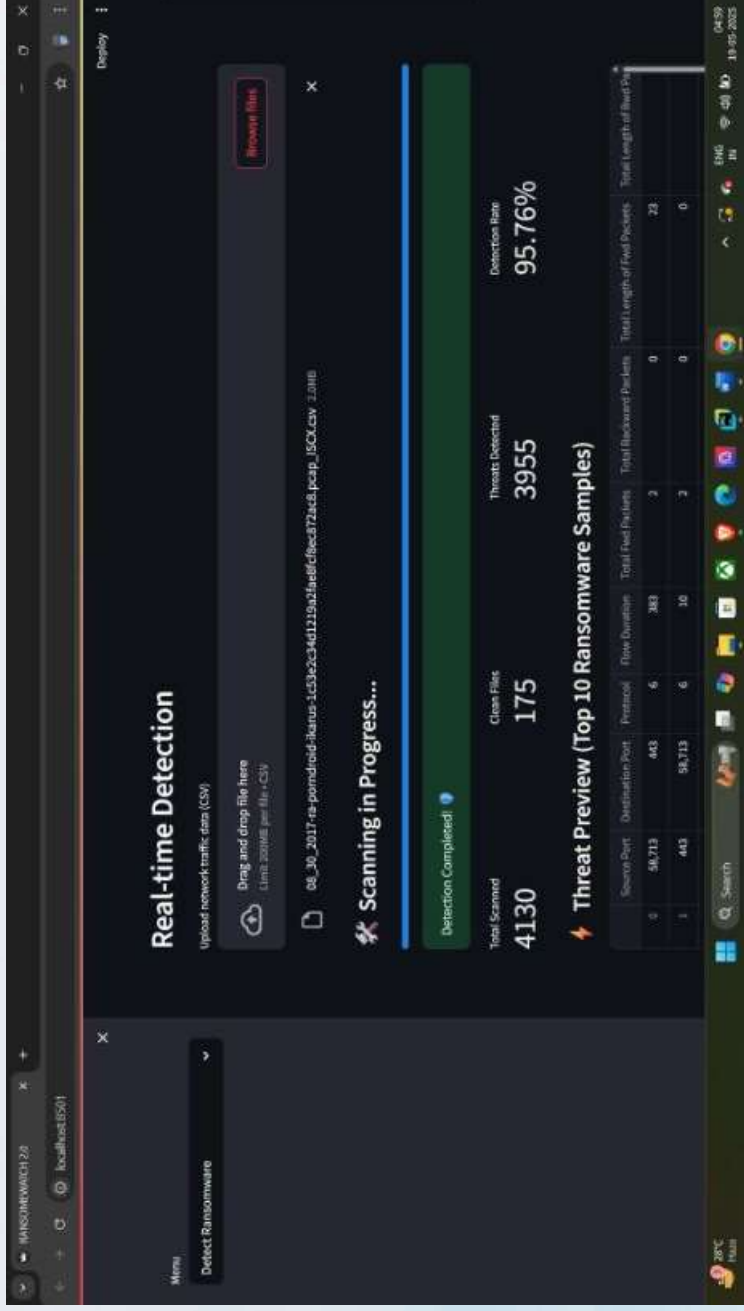


fig:Animated Windows-style progress bar demonstrating live feedback during model training or ransomware scanning.

Live Progress Animation

An animated Windows-style progress bar demonstrating real-time feedback during scanning processes, enhancing the visual status updates.



Screenshot of the
showing ransomware
summary metrics
samples for review

fig:User interface screenshot displaying ransomware detection summary metrics and top flagged samples

Streamlit User Interface

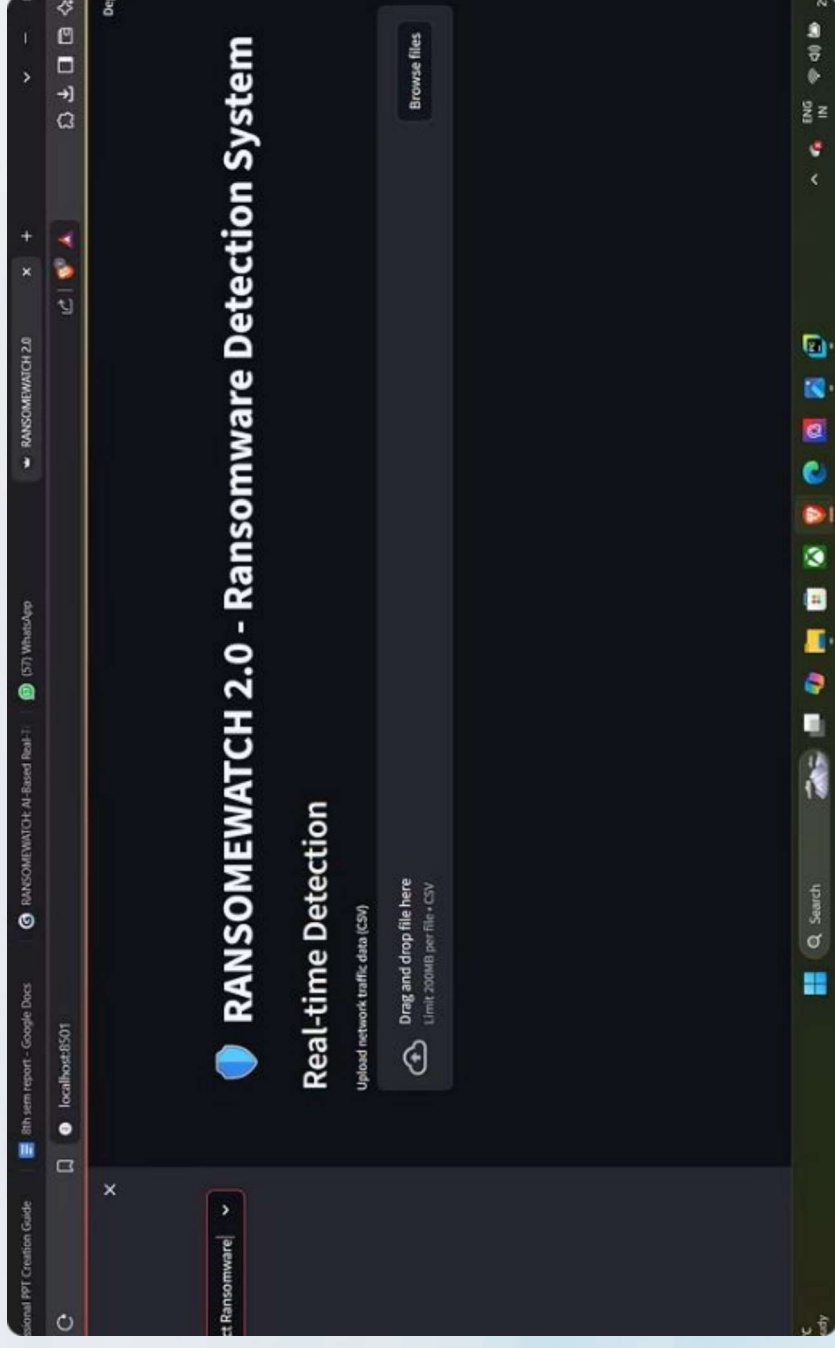


fig: Upload Network Traffic

Users submit network traffic logs in CSV format, including benign and suspicious samples, for detailed analysis.

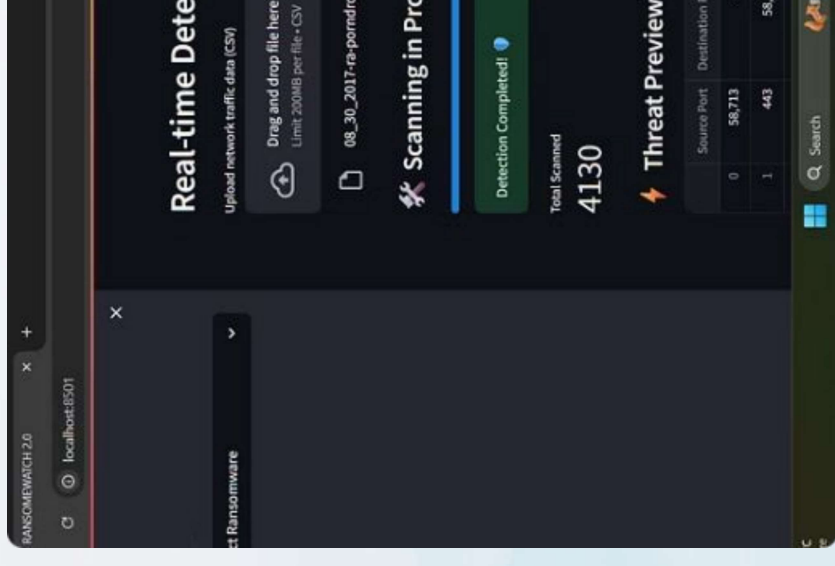


fig: Real-Time Rans

The system scans uploads and classifies threats with an in

Model Performance Metrics



Accuracy

Overall correct predictions.

98.5%



Precision

Minimising false positives.

99.1%



Recall

Minimising false negatives.

98.0%



AUC-ROC

High distinguishability.

0.99

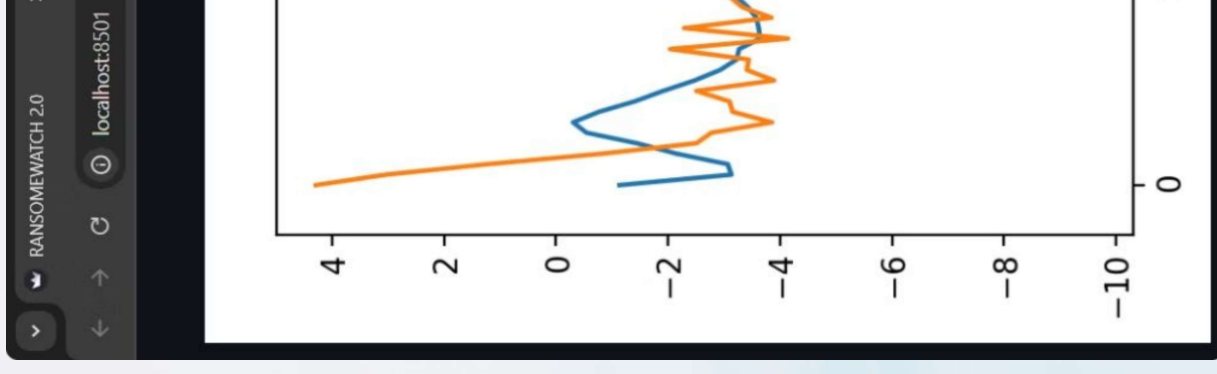


fig: Train

Model Performance Visualizations

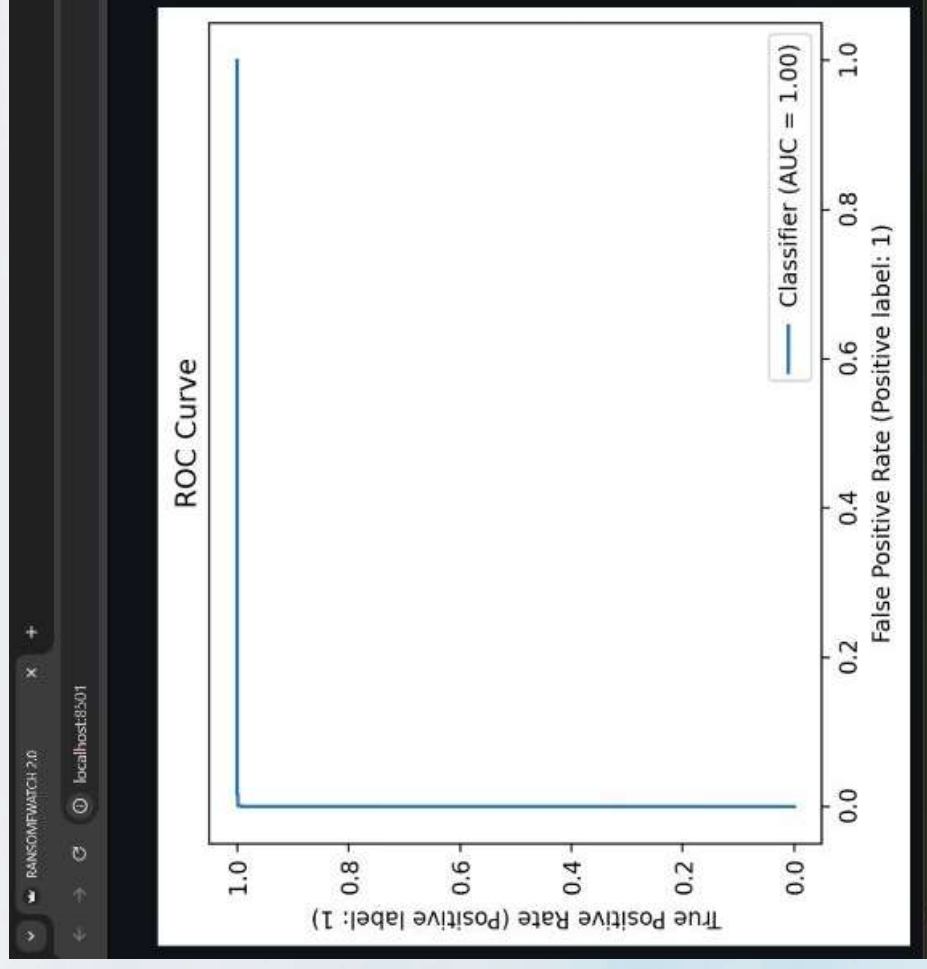


fig: ROC Curve

The Receiver Operating Characteristic (ROC) curve illustrates the trade-off between true positive rate and false positive rate across different classification thresholds, demonstrating the model's ability to distinguish between classes

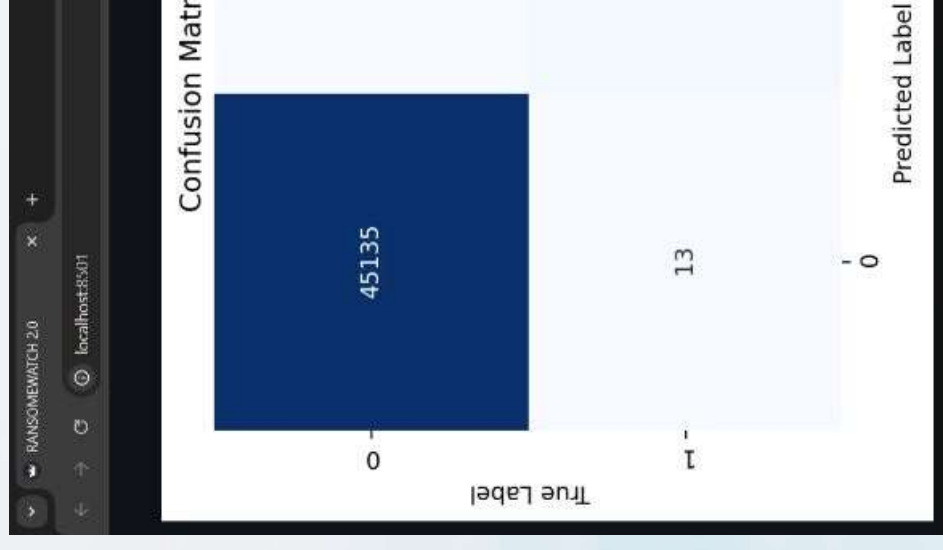


fig:Confusion Matrix

This matrix summarizes the model's performance, providing insight into classification accuracy, of true positives, true negatives,

Future Work and Enhancements

Dataset Expansion

Include newer ransomware families for broader coverage.

Real-Time Alerts

Integrate alert systems and auto-quarantine features.

Enterprise Integration

Connect with enterprise networks and cloud APIs.

UI Improvement

Enhance UI with advanced visual analytics dashboards.

Conclusion

RANSOMEWATCH uniquely combines advanced AI with real-time usability. It effectively tackles data imbalance, offering transparent and explainable outcomes. RANSOMEWATCH is ready for immediate deployment and further expansion within existing contexts.

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THANK YOU